



From CO₂ Emissions to Olefins

Reimagining the Carbon Loop

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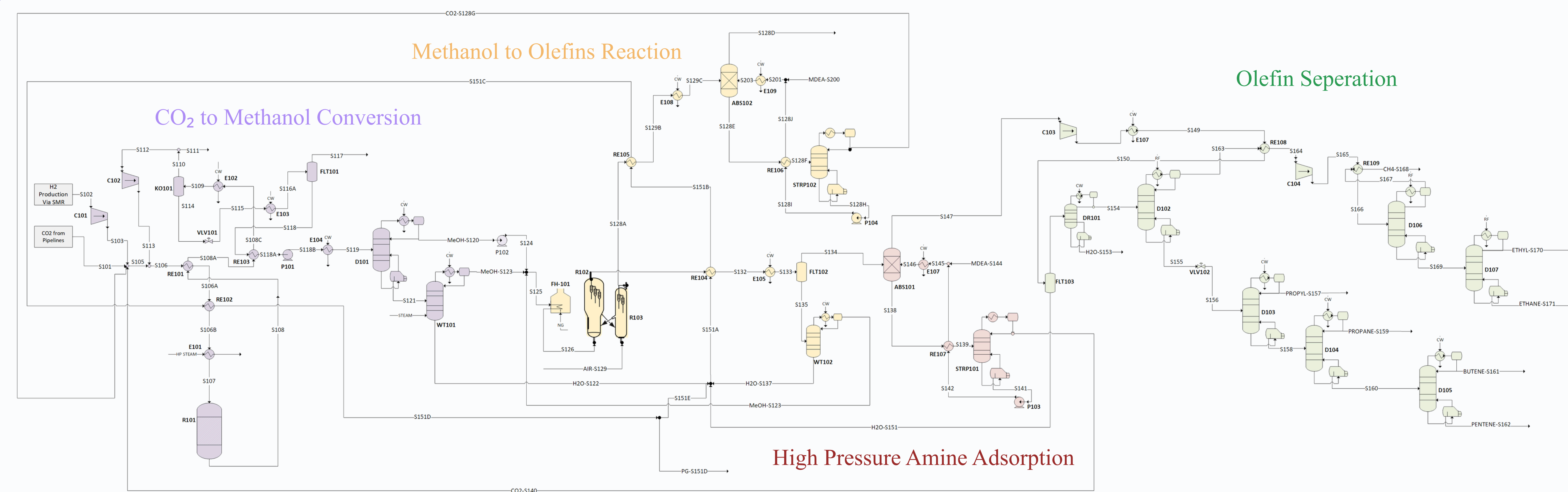
Background

Objective: Transform CO₂ from waste to valuable product to accelerate global decarbonization goals and provide a viable solution for industrial emitters.

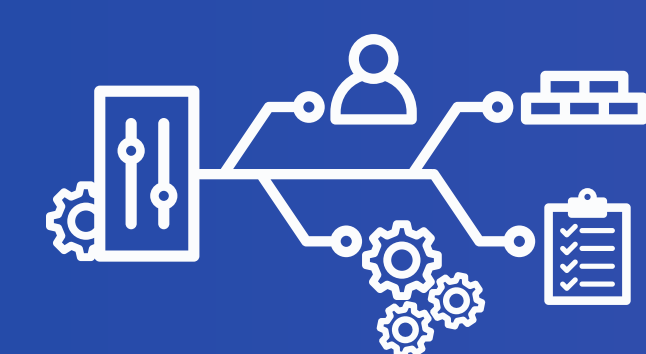
Our Solution:

- Convert CO₂ into methanol via catalytic hydrogenation.
- Upgrade methanol into polymer-grade olefins using MTO and advanced purification steps.
- Implement recycle loops to ensure zero carbon-emission goals.

Process Flow Diagram



Design Basis:

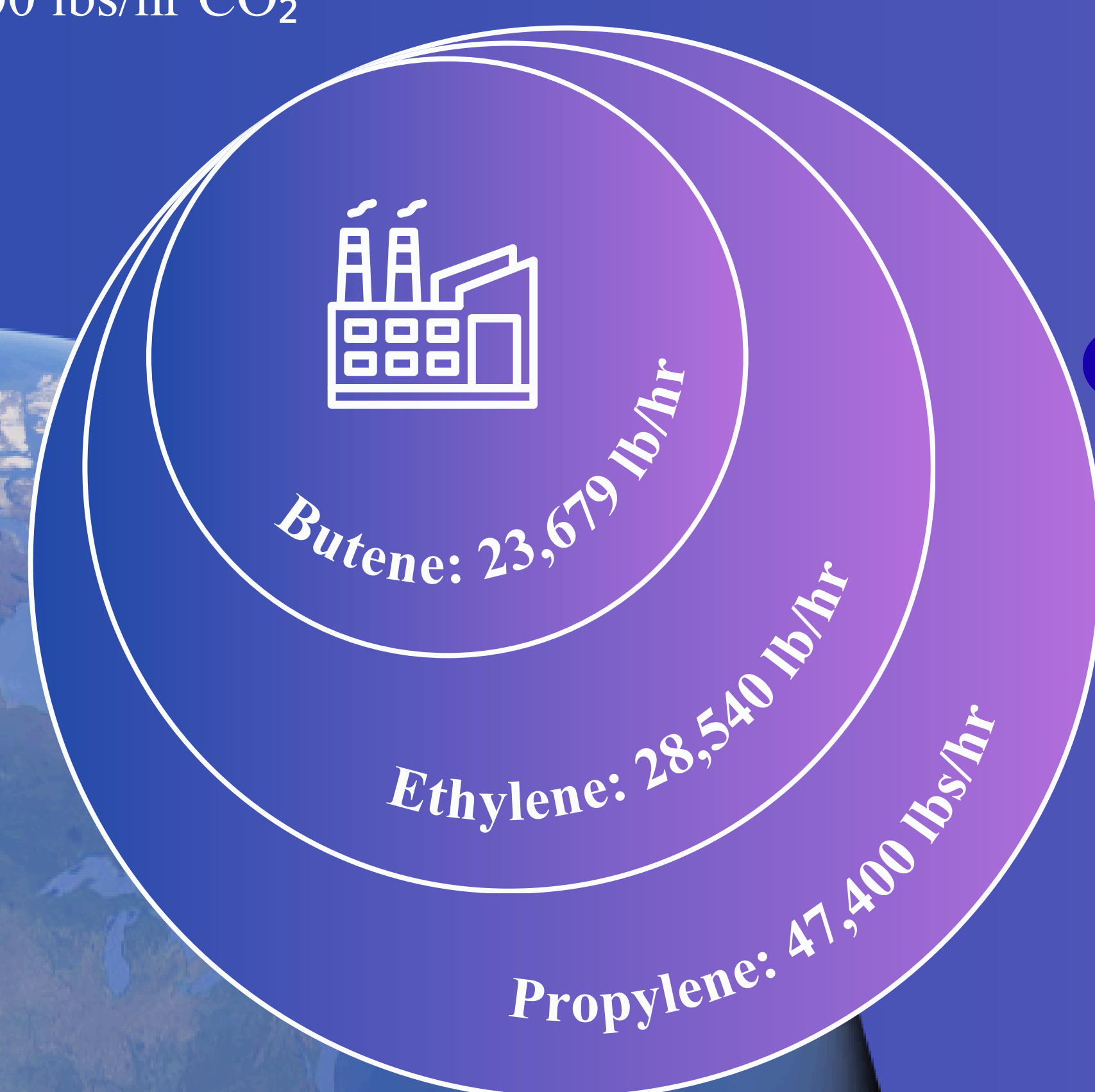


Consumables:

- Methyldiethanolamine
- Cu/ZnO/Al₂O₃ catalyst
- SAPO-34 catalyst

Plant Capacity: 388,100 lbs/hr CO₂ removed from emission

Location: Houston, TX

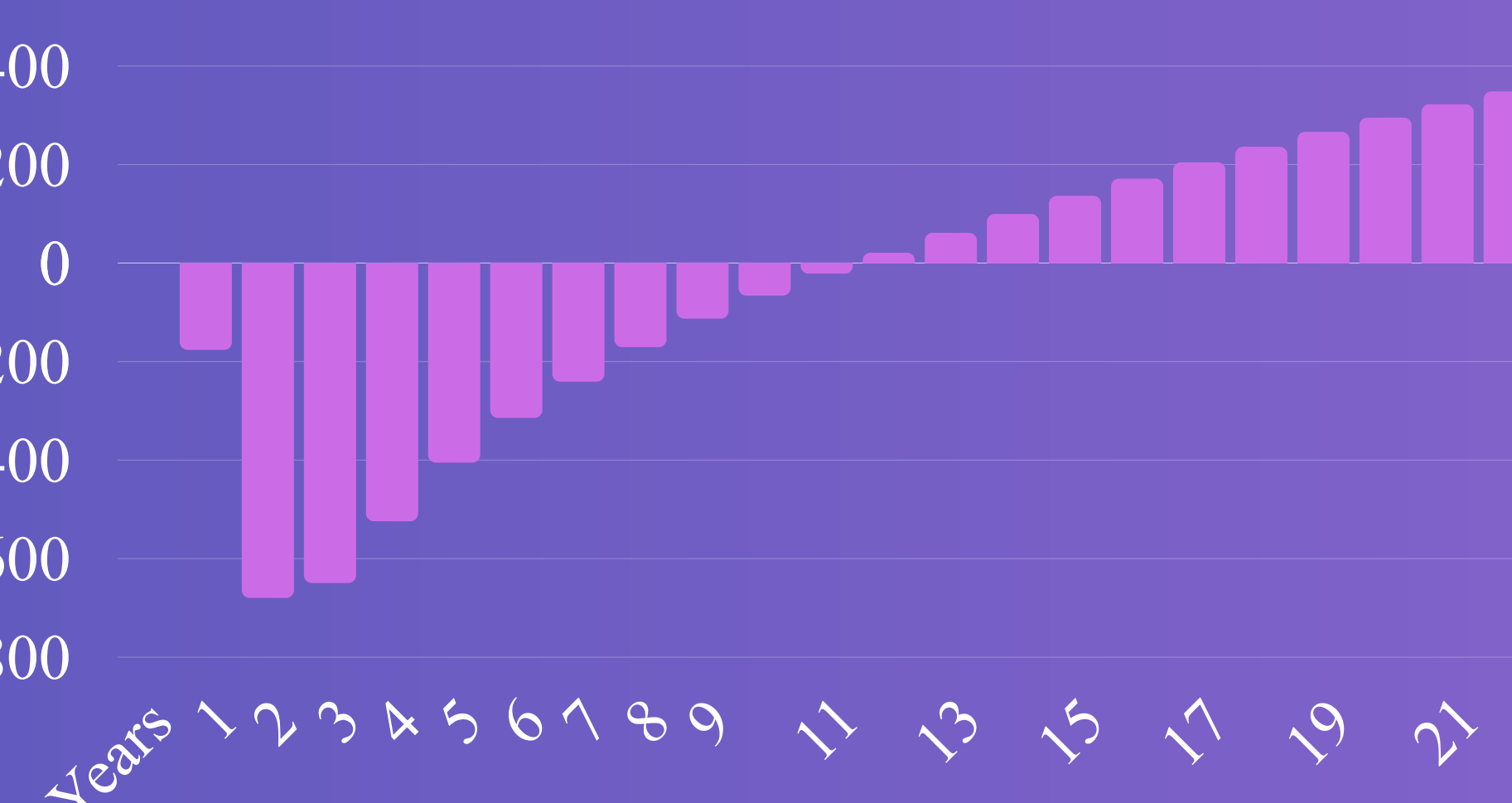


Economics

Costs Summary (\$MM):

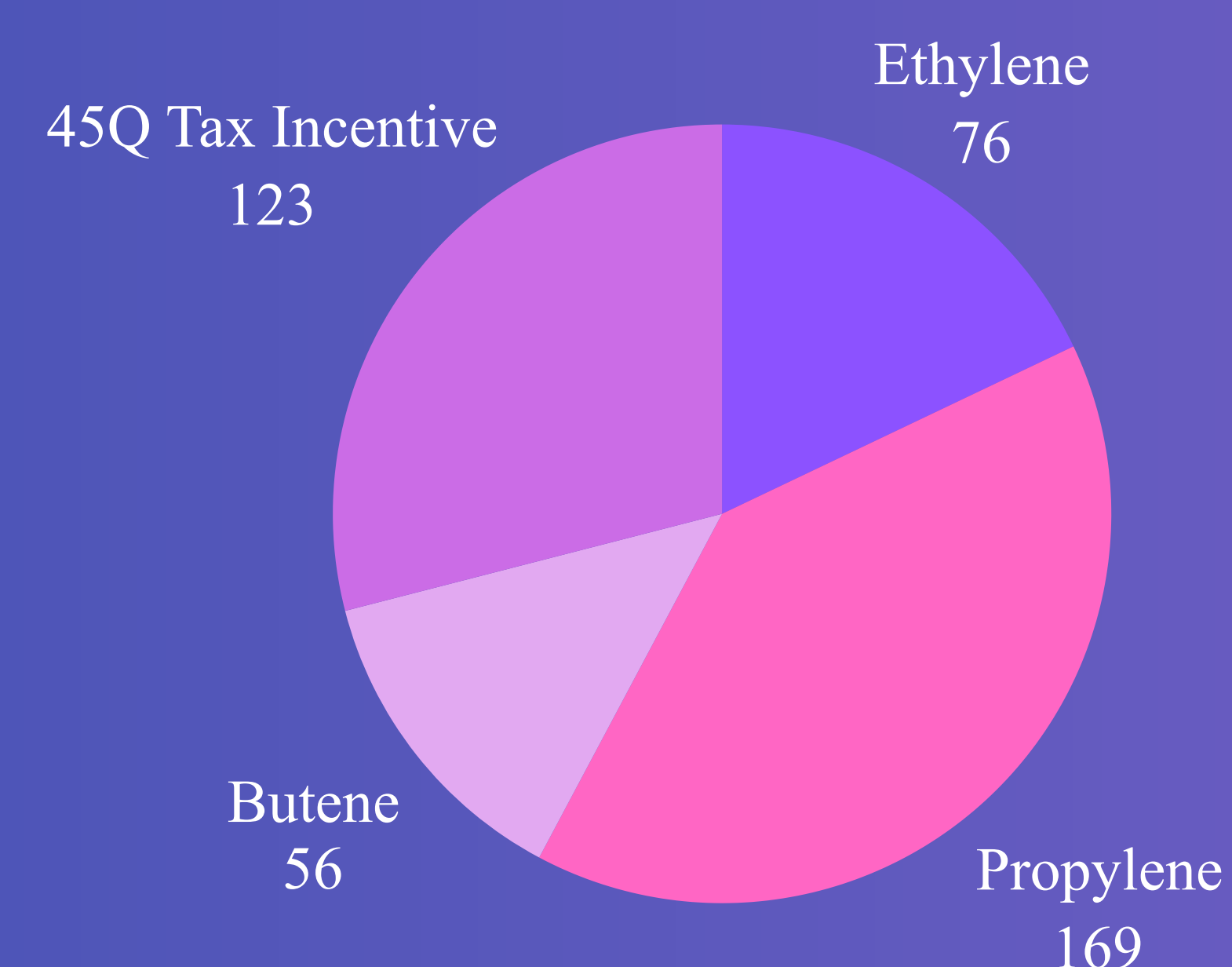
ISBL	423
Total Capital Costs	740
VCOP	260
FCOP	56
Revenue	424
Gross Profit	310

Net Present Value:



Revenue Breakdown (\$MM):

Summary:



- Economics of this process is heavily dependent on our capital intensive equipment.
- The profitability is dependent on the tax incentives.
- Our payback period is 9 years.
- Under current and predicted future economic conditions, the process poses as a viable solution for industrial emitters.

Safety

Process Safety: Risk of high-pressure and flammable hydrogen reactions in reactors → requires continuous monitoring, pressure relief systems, and emergency shutdowns to prevent explosions or runaway reactions.

Operational Safety: Worker exposure to extreme conditions and toxic chemicals (methanol, CO, cryogenic streams) → controlled through strict PPE use, insulation, and safe handling procedures.

Environmental Safety: Emission and release control of CO₂, VOCs, and wastewater → managed through recycling, controlled venting, and compliance with EPA regulations.

Recommendations



Focus on reducing costs and resource use by optimizing energy, water, hydrogen utilization, and reactor performance.



Shift from selling products to offering CO₂-reduction services, generating stable revenue through contracts and scalability.



Increase value by integrating polymer production, expanding strategically, and optimizing feedstock and production costs.